

RobustGaSP

1D test example

We have a 1-dimensional input, one output, and 20 noise-free observations, obtained using the function $f(x) = 2 \sin(3x - 0.5)$:

$x = \{0.1, 0.1421053, 0.1842105, 0.2263158, 0.2684211, 0.3105263, 0.3526316, 0.3947368, 0.4368421, 0.4789474, 0.5210526, 0.5631579, 0.6052632, 0.6473684, 0.6894737, 0.7315789, 0.7736842, 0.8157895, 0.8578947, 0.9000000\}$

$y = \{-0.4, -0.15, 0.11, 0.36, 0.6, 0.84, 1.06, 1.26, 1.45, 1.61, 1.75, 1.86, 1.94, 1.98, 2, 1.98, 1.94, 1.86, 1.75, 1.62\}$

which is plotted in Figure 1.

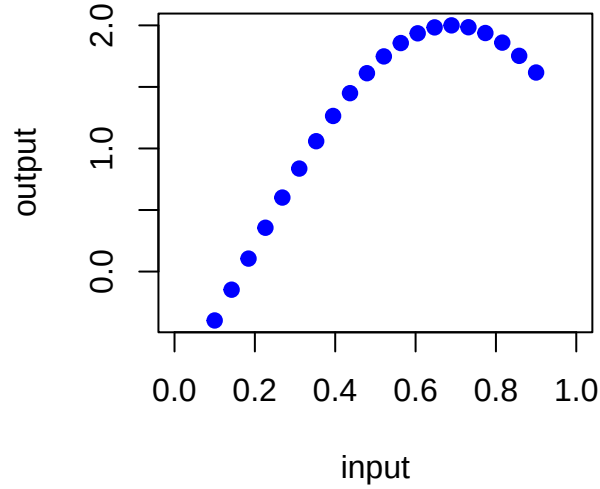


Figure 1: 1D example observations

Going forward, we will assume a linear mean function of the form

$$m(x) = \beta_0 + \beta_1 x.$$

In RobustGaSP's `rgasp()` function, this involves specifying the `trend` matrix H :

```
##      [,1]      [,2]
## [1,]    1 0.1000000
## [2,]    1 0.1421053
## [3,]    1 0.1842105
```

```
## [4,] 1 0.2263158
## [5,] 1 0.2684211
## [6,] 1 0.3105263
## [7,] 1 0.3526316
## [8,] 1 0.3947368
## [9,] 1 0.4368421
## [10,] 1 0.4789474
## [11,] 1 0.5210526
## [12,] 1 0.5631579
## [13,] 1 0.6052632
## [14,] 1 0.6473684
## [15,] 1 0.6894737
## [16,] 1 0.7315789
## [17,] 1 0.7736842
## [18,] 1 0.8157895
## [19,] 1 0.8578947
## [20,] 1 0.9000000
```

as well as setting `zero.mean` equal to "No". The Gauss (squared exponential) *and* Matern 5/2 kernels are compared, leading to the creating of four functions, `gp_km_gauss()`, `gp_km_matern52()`, `gp_rgasp_gauss()` and `gp_rgasp_matern52()`, reprinted below:

```
# gp_km_gauss <- function(input_matrix, output_matrix){
#   km(~.,
#     input_matrix,
#     output_matrix,
#     covtype="gauss",
#     optim.method="BFGS",
#     control=list(maxit=500)
#   )
# }

# gp_km_matern52 <- function(input_matrix, output_matrix){
#   km(~.,
#     input_matrix,
#     output_matrix,
#     covtype="matern5_2",
#     optim.method="BFGS",
#     control=list(maxit=500)
#   )
# }

gp_rgasp_gauss <- function(input_matrix, output_matrix, H_matrix){
  rgasp(design = input_matrix,
    response = output_matrix,
    trend = H_matrix,
    lower_bound=FALSE,
    kernel_type = 'pow_exp', alpha = rep(2, nrow(input_matrix)),
    zero.mean = "No"
  )
}

gp_rgasp_matern52 <- function(input_matrix, output_matrix, H_matrix){
  rgasp(design = input_matrix,
```

```

    response = output_matrix,
    trend = H_matrix,
    lower_bound=FALSE,
    zero.mean = "No"
)
}

```

Note that `lower_bound` is set to `FALSE` - Jonathan doesn't *think* this is incorrect.

The output from the two RobustGaSP functions is shown below (first for the Gauss kernel and second for the Matern 5/2).

```

## The upper bounds of the range parameters are Inf
## The initial values of range parameters are 0.001124683
## Start of the optimization 1 :
## The number of iterations is 9
## The value of the marginal posterior function is -10.03534
## Optimized range parameters are 0.01213395
## Optimized nugget parameter is 0
## Convergence: TRUE
## The initial values of range parameters are 0.004
## Start of the optimization 2 :
## The number of iterations is 9
## The value of the marginal posterior function is -10.03534
## Optimized range parameters are 0.01213395
## Optimized nugget parameter is 0
## Convergence: TRUE

## The upper bounds of the range parameters are Inf
## The initial values of range parameters are 0.001124683
## Start of the optimization 1 :
## The number of iterations is 40
## The value of the marginal posterior function is 87.31155
## Optimized range parameters are 17.12366
## Optimized nugget parameter is 0
## Convergence: FALSE
## The initial values of range parameters are 0.004
## Start of the optimization 2 :
## The number of iterations is 30
## The value of the marginal posterior function is 5.881201
## Optimized range parameters are 535952901783
## Optimized nugget parameter is 0
## Convergence: TRUE

```

We believe there are two optimisations since the first is getting a starting point for the second. Jonathan doesn't think it's a problem that we sometimes get a `FALSE` result for `Convergence`.

To visualise the posterior mean functions returned by these four functions, predictions are made at 1000 new points on the range $[0, 1]$. Once again, `rgasp` requires specification of the 'trend' matrix H , this time containing the testing inputs.

These 1000 points are plotted in Figure 2 (red line), along with 95% credible intervals (red dashed line) along with the actual function (black line) and 20 observations (blue points).

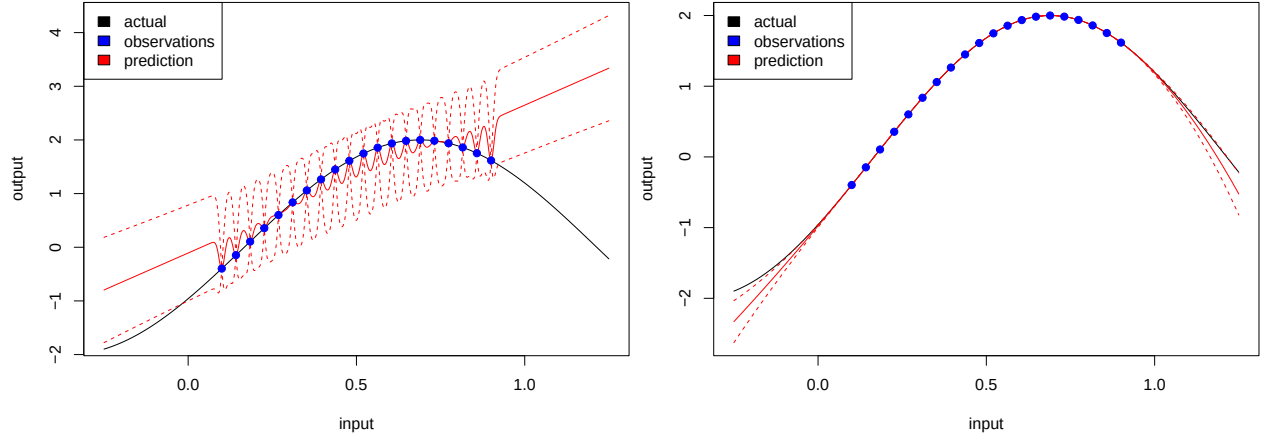


Figure 2: Top: DiceKriging with Gauss kernel (left) and Matern 5/2 (right). Bottom: RobustGaSP with Gauss kernel (left) and Matern 5/2 (right).